

Technical Document #107: Retrieval Augmented Generation - Comprehensive Overview

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Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Hybrid search combines keyword-based and semantic search.
- Hallucination detection identifies when LLMs generate factually incorrect information.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.